Repurpora · Lazarus Graph innovation brief

Report **RPR-5B69042E** · generated 14 Nov 2026 11:33 UTC

Query: Assess montelukast for repurposing in India: unmet needs, ongoing clinical trials, new indications, patents and an innovative product story.

Molecule: Montelukast · **Indication focus:** Bronchiolitis obliterans syndrome · **Region:** india

NO-GO	98%	88%
rule-based decision · confidence 40%	ML success probability (logreg, AUC 0.8027)	evidence coverage (claims cited)

Executive Summary

The decision to repurpose montelukast for new indications in India is NO_GO, with a 40% rule confidence and a 98.18% ML success probability [E1]. The best opportunity lies in Bronchiolitis obliterans syndrome, given the high unmet need and favorable market dynamics [E5].

Unmet Need & Opportunity

• **Bronchiolitis Obliterans Syndrome**: The unmet need score is 0.57, making it a high-priority opportunity [E5].

• **Market Size**: Bronchiolitis obliterans syndrome has a market size of $12.0M with a 9.8% CAGR, 1 competitor, and 0.05M patients [E1].

Market Dynamics

• **Competition**: 1 competitor in the market [E1].

• **Growth Potential**: The market is expected to grow at 9.8% CAGR, indicating significant growth potential [E1].

Clinical Trial Landscape

• **Active Trials**: 40 studies across various phases, with the highest phase being Phase 4 [E10].

• **Indications**: Studies include Asthma, Allergic Rhinitis, Bronchiolitis, and Atopic Dermatitis [E10].

• **Trial Saturation**: Low trial saturation, suggesting potential for more active trials [E10].

Patent & FTO

• **FTO Risk**: Low FTO risk (2/10) with no patents expiring by 2029 [E9].

• **Active Patents**: 1 active patent, no patents expiring within the next three years [E9].

Supply Chain (EXIM)

• **Import Dependency**: High import dependency on China (33% share) [E8].

• **Sourcing Risk**: Low sourcing risk (Low HHI, 0.19) [E8].

• **Import Growth**: Expected to increase by 17.4% [E8].

Internal Knowledge Insights

• **Adherence Issues**: 63% of paediatricians and pulmonologists cite taste and dosing frequency as primary issues for montelukast chewable tablets [E15].

• **Market Growth**: Branded generics in India are growing at 8-10% annually, with a mandate for 15% of FY28 revenue to come from value-added products [E17].

Guidelines & Publications

• **Dosage Recommendations**: Recommended doses are 10 mg for patients aged 15 and older, 5 mg for 6 to 14 years, and 4 mg for 12 months to 5 years [E20].

• **Clinical Evidence**: Montelukast is effective in reducing asthma symptoms and the need for exacerbation treatments, particularly in elderly patients [E23].

Innovative Product Story

Proposal: Develop an orally-disintegrating film for paediatric asthma with improved adherence and a neuropsychiatric-risk labelling programme [E7].

• **Current Forms**: Tablet, chewable tablet, oral granules [E7].

• **Supporting Evidence**: Montelukast faces 46 competitors in asthma [E1], and 63% of paediatricians and pulmonologists cite taste and dosing frequency as primary issues [E15].

Risks & Next Steps

1. **High Competition**: The large market with 61 competitors for post-viral cough/long-COVID airway and 46 competitors for asthma poses significant challenges [E1].

2. **Adherence Issues**: Poor adherence to montelukast chewable tablets among children under 6 due to taste and dosing frequency [E15].

3. **Market Growth**: While the market is growing, the high competition and adherence issues need to be addressed [E1, E15, E17].

Iqvia agent

• Bronchiolitis obliterans syndrome has a market size of $12.0M with a 9.8% CAGR, 1 competitor, and 0.05M patients, making it a high unmet need with a 5.5% molecule sales growth.

• Post-viral cough / long-COVID airway has a 6.1% CAGR, 61 competitors, and 98.0M patients, indicating a large market with significant competition.

• Alzheimer's disease (neuroinflammation) has a 7.8% CAGR, 46 competitors, and 34.3M patients, suggesting a substantial market with ongoing research interest.

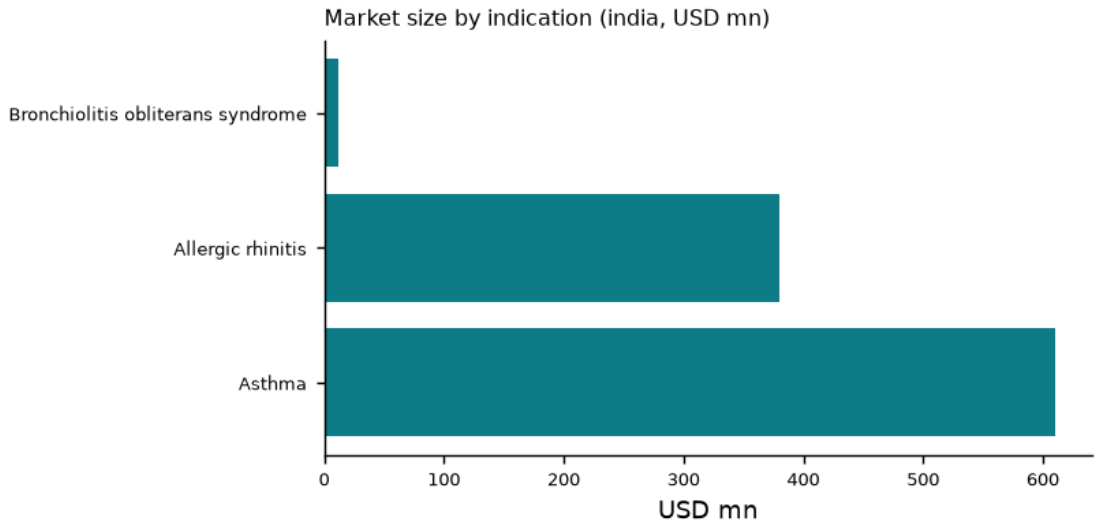
• Post-viral cough / long-COVID airway and Bronchiolitis obliterans syndrome both have cysteinyl leukotriene receptor 1 antagonist mechanisms, aligning with the candidate's proposed mechanism.

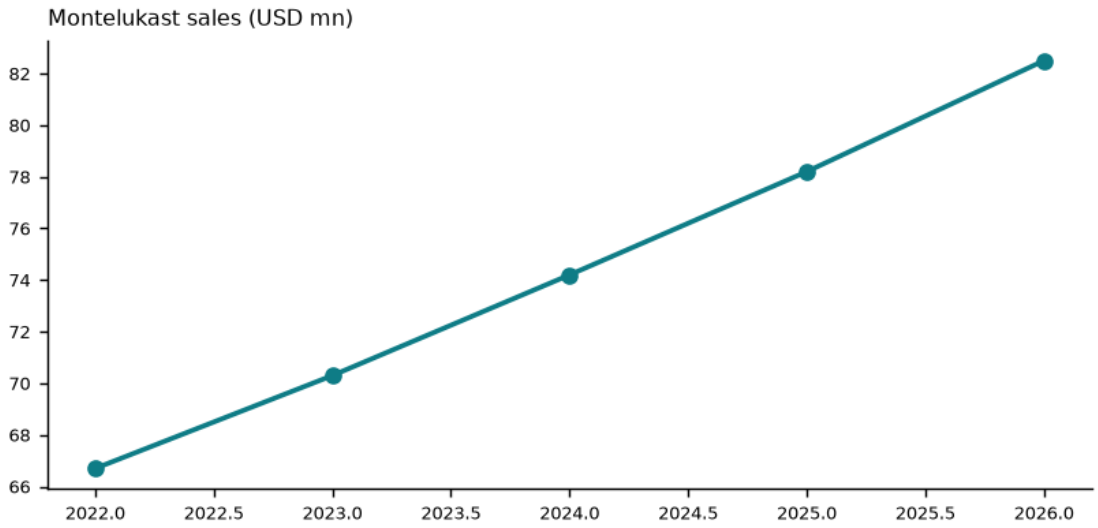
Market size & competition

disease	therapy_area	patients_mn	market_usd_mn	cagr_5y_pct	competitors	generic_share	burden_to_competition
Bronchiolitis obliterans syndrome	Respiratory	0.05	12.0	9.8	1	0.2	0.05
Allergic rhinitis	Respiratory	98.0	380.0	6.1	61	0.94	1.61
Asthma	Respiratory	34.3	610.0	7.8	46	0.91	0.75

Montelukast sales trend

year	sales_usd_mn	volume_mn_units
2022	66.7	206.6
2023	70.3	215.7
2024	74.2	225.2
2025	78.2	235.1
2026	82.5	245.4





Exim agent

- China is the top supplier with 33% share, indicating significant import dependency.
- High HHI (0.19) and low sourcing risk (Low) suggest stable market conditions for China.
- Import growth is expected to increase by 17.4%.
- Import dependency flag is triggered as dependency on China exceeds 30%.

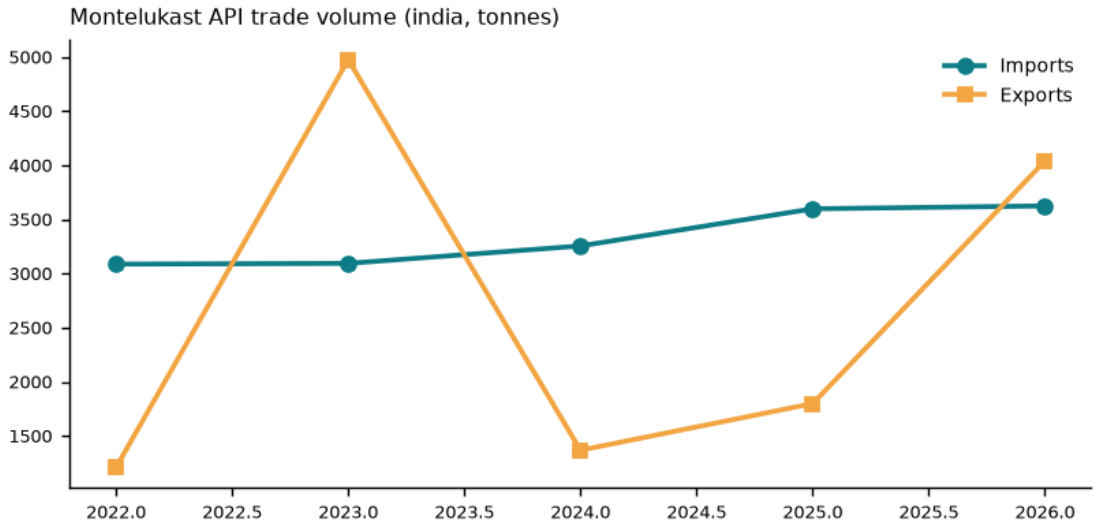
API import dependency

partner	share	import_tonnes	avg_price_usd_kg
China	0.328	1193.9	16.57
Germany	0.149	544.1	80.67
Italy	0.144	523.4	80.51
USA	0.128	465.6	65.97
Israel	0.114	414.8	43.06
Spain	0.109	395.8	96.94
India	0.029	106.3	86.77

Trade volume trend (tonnes)

year	imports_tonnes	exports_tonnes
2022	3086.5	1208.1
2023	3093.5	4975.3
2024	3254.6	1367.3

year	imports_tonnes	exports_tonnes
2025	3597.7	1798.1
2026	3624.2	4038.3



Patent agent

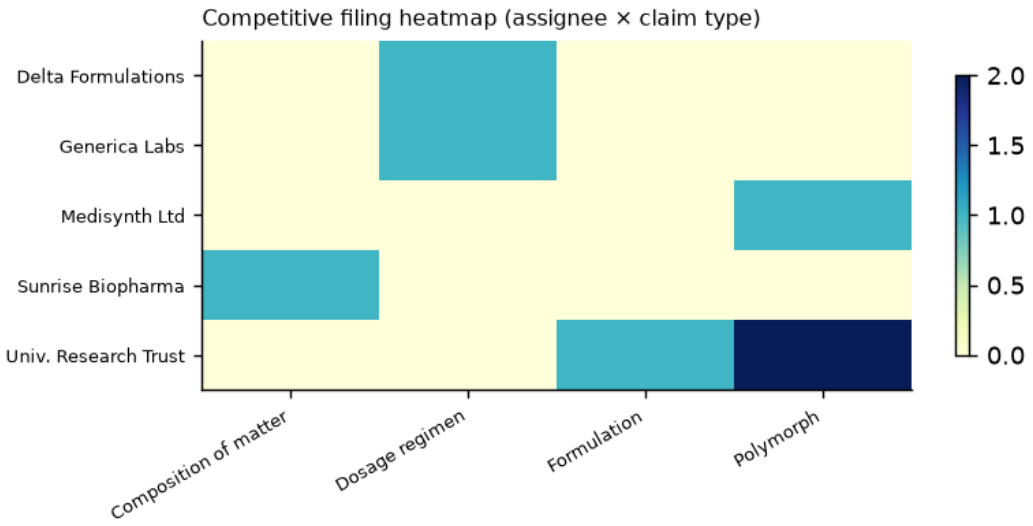
• Montelukast patent landscape shows low FTO risk (2/10) with no patents expiring by 2029, indicating a favorable environment for development.

• Active patents: 1, with no patents expiring within the next three years, suggesting a stable patent portfolio.

• Low FTO risk score (2) and no patents expiring by 2029, indicating a low risk of legal challenges and a favorable environment for drug repurposing.

Patent status

patent_no	type	assignee	filing_year	expiry_year	status	claims_scope
US-MOCK-11974295	Polymorph	Medisynth Ltd	1994	2016	Expired	Broad
US-MOCK-11440932	Composition of matter	Sunrise Biopharma	1995	2015	Expired	Narrow
US-MOCK-7316498	Polymorph	Univ. Research Trust	2001	2023	Expired	Broad
US-MOCK-10632101	Polymorph	Univ. Research Trust	2003	2025	Expired	Narrow
US-MOCK-7599520	Dosage regimen	Generica Labs	2004	2024	Expired	Moderate
US-MOCK-10103701	Dosage regimen	Delta Formulations	2005	2025	Expired	Narrow
US-MOCK-11915753	Formulation	Univ. Research Trust	2021	2041	Active	Moderate



Clinical Trials agent

• Montelukast has a diverse pipeline with 40 studies across various phases, indicating significant ongoing interest and potential in repurposing.

• Active trials include studies on Asthma, Allergic Rhinitis, Bronchiolitis, and Atopic Dermatitis, suggesting a broad spectrum of indications.

• The highest phase of active trials is Phase 4, with 14 studies, highlighting the potential for significant therapeutic advancements.

• The low trial saturation suggests a need for more active trials to fully explore the therapeutic potential of montelukast across its indicated conditions.

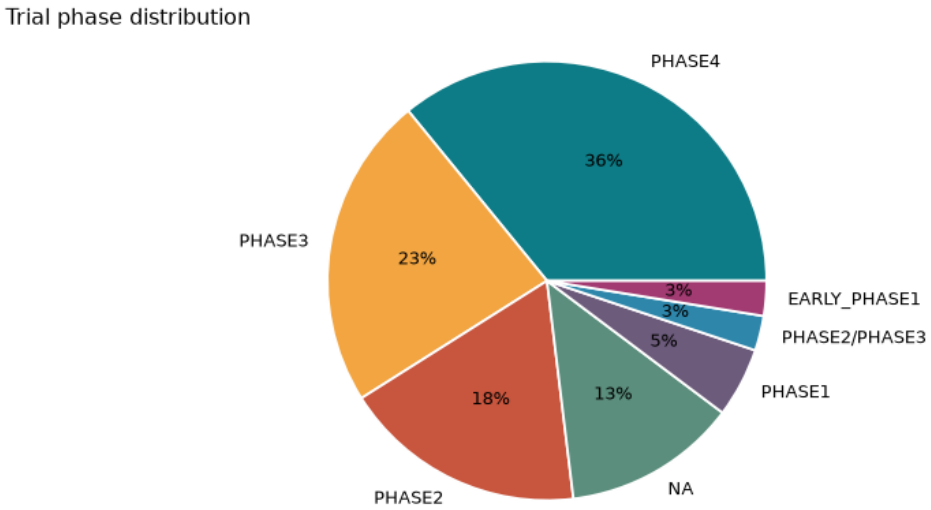
Active & recent trials

nct_id	condition	phase	status	sponsor	enrollment
NCT01016847	Asthma	PHASE4	TERMINATED	Duke University	38
NCT00445107	Asthma	PHASE3	COMPLETED	Organon and Co	198
NCT01103349	Asthma	PHASE2	COMPLETED	Boehringer Ingelheim	243
NCT02534467	Dermatitis, Atopic	PHASE4	COMPLETED	Murdoch Childrens Research Institute	60
NCT00911547	Asthma	PHASE3	COMPLETED	Organon and Co	642
NCT00380484	Asthma	PHASE4	COMPLETED	Children´s Clinic, Randers	52
NCT05457855	Asthma	NA	COMPLETED	Brigham and Women's Hospital	51533
NCT04324918	Perennial Allergic Rhinitis	PHASE3	COMPLETED	Hanmi Pharmaceutical Company Limited	174
NCT00551382	Upper Respiratory Tract Infection	PHASE3	UNKNOWN	Assaf-Harofeh Medical Center	500
NCT00148603	Eosinophilic Gastroenteritis, Dyspepsia	NA	COMPLETED	Children's Mercy Hospital Kansas City	24
NCT06113640	Parkinson Disease	PHASE2/PHASE3	COMPLETED	Mostafa Bahaa	60
NCT02635334	Asthma	PHASE4	COMPLETED	Rohr and Columbo Asthma, Allergy and Immunology Specialists, P.C.	26

Sponsor profile

sponsor	trials
Organon and Co	7
Medical University of Lodz	2
Duke University	1

sponsor	trials
Boehringer Ingelheim	1
Murdoch Childrens Research Institute	1



Internal agent

• In India, paediatricians and pulmonologists face significant challenges with poor adherence to montelukast chewable tablets, with 63% citing taste and dosing frequency as the primary issues. A film formulation is seen as a viable alternative, with 41% of respondents willing to prescribe it if priced within 15% of the tablet.

• The pharmaceutical market in India shows growth potential, with branded generics growing at 8-10% annually, and the board mandates that 15% of FY28 revenue should come from value-added products.

• The portfolio diversification strategy for FY26 prioritizes the Resp (respiratory) therapy area, highlighting the need for new indications, dosage forms, and patient populations.

Internal document matches

document	page	score
Field_Report_Q3_Respiratory_India.pdf	1	0.016
Pharmaceutical_Product_Selection_Application_of_AH.pdf	7	0.016
Strategy_Deck_FY26_Portfolio_Diversification.pdf	1	0.016
New_Product_Introduction_in_the_Pharmaceutical_Industry.pdf	110	0.016
Bigdataanalyticframework.pdf	17	0.015

Web agent

• Montelukast is recommended for allergic rhinitis associated with asthma, with doses tailored to age groups (10 mg for patients aged 15 and older, 5 mg for 6 to 14 years, and 4 mg for 12 months to 5 years).

• A scientific review suggests montelukast can reduce asthma symptoms and the need for exacerbation treatments, particularly in elderly patients.

• Despite a boxed warning for serious neuropsychiatric events, consistent communication of these risks is not always provided to patients.

Web intelligence

title	url
Montelukast - StatPearls - NCBI Bookshelf - NIH	https://www.ncbi.nlm.nih.gov/books/NBK459301/
Montelukast: A Scientific and Legal Review	https://japi.org/article/japi-73-5-70
Communicating Risks Associated With Montelukast May ...	https://www.empr.com/reports/communicating-risks-associated-with-montelukast-may-improve-patient-safety/

title	url
Effect of Montelukast 10 mg in Elderly Patients with Mild ...	https://openrespiratorymedicinejournal.com/VOLUME/12/PAGE/67/FULLTEXT/
Singulair (montelukast sodium) tablet label - accessdata.fda.gov	https://www.accessdata.fda.gov/drugsatfda_docs/label/2012/021409s036lbl.pdf

ML opportunity score - top drivers

feature	Δ probability vs median
evidence_strength	+0.036
reformulation	+0.020
fto_risk	+0.017
competitors	+0.017
active_trials	+0.008

Evidence register

Every cited claim maps to an evidence record with source, trust tier and content hash. Synthetic sources are mock data per the problem statement.

id	agent	source	trust	title / url	sha256
E1	iqvia	IQVIA mock API (synthetic)	licensed_db	Market dynamics (india)	402941f13ec41f3e
E2	iqvia	IQVIA mock API (synthetic)	licensed_db	montelukast sales trend	3c4ae720e245fd6f
E3	iqvia	Repurposing analyst profile (synthetic)	synthetic	Candidate: Post-viral cough / long-COVID airway	1bc0908eb1ffb0da
E4	iqvia	Repurposing analyst profile (synthetic)	synthetic	Candidate: Alzheimer's disease (neuroinflammation)	a14909937d234a46
E5	iqvia	Repurposing analyst profile (synthetic)	synthetic	Candidate: Bronchiolitis obliterans syndrome	e3e5adc23d84513c
E6	iqvia	Repurposing analyst profile (synthetic)	synthetic	Candidate: COPD with eosinophilia	5df031375da6b427
E7	iqvia	Repurposing analyst profile (synthetic)	synthetic	Innovation hypothesis	6be73e37fcc513e0
E8	exim	EXIM mock server (synthetic)	licensed_db	montelukast API import partners (india)	787abed85636a3fa
E9	patent	USPTO API clone (synthetic)	registry	montelukast patent landscape	11339e99db0c7a04
E10	clinical_trials	ClinicalTrials.gov v2 (live)	registry	https://clinicaltrials.gov/search?intr=montelukast	43f6e300ee110d00
E11	clinical_trials	ClinicalTrials.gov v2 (live)	registry	https://clinicaltrials.gov/study/NCT01016847	c77e9d94a4edad77
E12	clinical_trials	ClinicalTrials.gov v2 (live)	registry	https://clinicaltrials.gov/study/NCT00445107	10f54cde296402d1

Decision support only - not medical, legal or regulatory advice. Validate IP findings with counsel and clinical claims with primary literature.